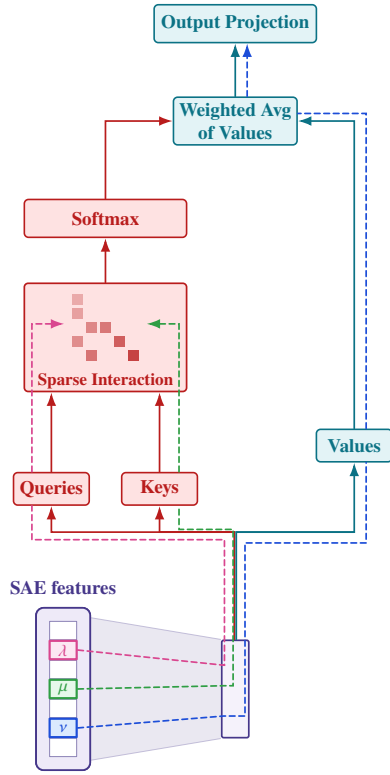
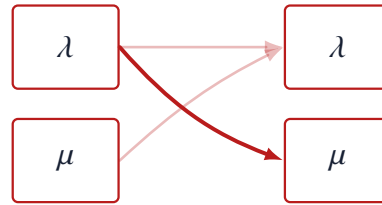


FRA Attribution

Decompose the head: each computation step is rewritten as a sparse tensor over SAE features. Attribute by magnitude.

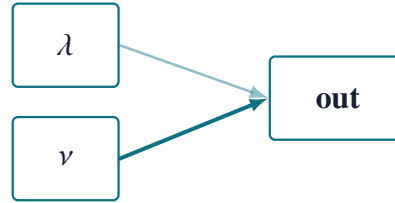


k pos



q pos

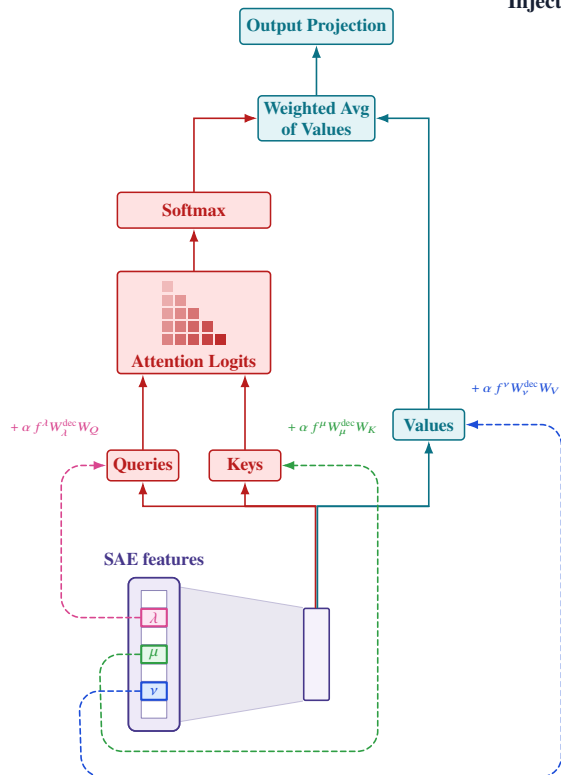
$$\text{FRA}_{qk,\lambda,\mu}^{\text{QK}} = \frac{f_q^\lambda f_k^\mu}{\sqrt{d_h}} W_\lambda^{\text{dec}} W_{QK} (W_\mu^{\text{dec}})^\top$$



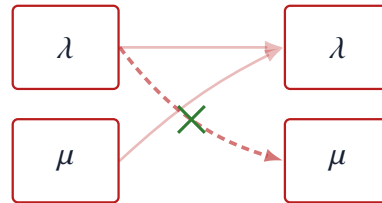
$$\text{FRA}_{qk,v}^{\text{OV}} = A_{qk} f_k^\nu \|W_\nu^{\text{dec}} W_{OV}\|_2$$

Targeted Intervention

Inject a single SAE feature direction into the head at the Q, K, or V site, scaled by α .

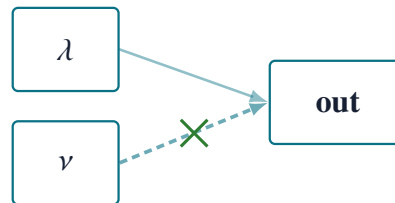


k pos



q pos

$$\begin{aligned} \tilde{Q}_q &= Q_q + \alpha f_q^\lambda W_\lambda^{\text{dec}} W_Q \\ \tilde{K}_k &= K_k + \alpha f_k^\mu W_\mu^{\text{dec}} W_K \end{aligned}$$



$$\tilde{V}_k = V_k + \alpha f_k^\nu W_\nu^{\text{dec}} W_V$$